One Contrastive Objective and Separate Temporal Uncertainty Training

This note explains the weighted supervised contrastive objective and the heteroscedastic temporal uncertainty objective in the supplied Gaussian contrastive implementation. The two uncertainty-aware weighting mechanisms modify a *single* contrastive objective, while a separate loss trains the variance estimator.

1 What the contrastive objective optimizes

For each video i, let $\boldsymbol{\mu}_i \in \mathbb{R}^{D_z}$ denote its learned semantic mean and $\boldsymbol{v}_i \in \mathbb{R}^{D_z}_{>0}$ its diagonal variance. Let s_{ij} be the cosine similarity of the two means and $\tau > 0$ the contrastive temperature. The positive set $\mathcal{P}_i$ contains other same-class representations; the negative set $\mathcal{N}_i$ contains different-class representations. Self-pairs are excluded. Define

$$Z_i = \sum_{a \in \mathcal{P}_i} \exp(s_{ia}/\tau) + \sum_{k \in \mathcal{N}_i} \alpha_{ik} \exp(s_{ik}/\tau). \tag{1}$$

The anchor objective is

$$\boxed{\ell_i = -\frac{1}{\sum_{j \in \mathcal{P}_i} w^+_{ij}} \sum_{j \in \mathcal{P}_i} w^+_{ij} \log \frac{\exp(s_{ij}/\tau)}{\displaystyle\sum_{a \in \mathcal{P}_i} \exp(s_{ia}/\tau) + \sum_{k \in \mathcal{N}_i} \alpha_{ik} \exp(s_{ik}/\tau)}.} \tag{2}$$

Let $V = \{i : |\mathcal{P}_i| > 0\}$ be the valid-anchor set. Then

$$\boxed{L_{\text{contrast}} = \frac{1}{|V|} \sum_{i \in V} \ell_i.} \tag{3}$$

All positive log-probabilities for anchor i share the same denominator Z_i.

2 Role of the positive weights

The coefficient w^+_{ij} controls the contribution of each positive log-probability. Augmented views from the same original video receive weight one; other same-class positives receive curriculum-dependent weights. The division by $\sum_j w^+_{ij}$ makes the anchor objective a weighted *average*, not a sum that grows automatically with the number or total weight of positives.

For diagonal Gaussian statistics, define the uncertainty-normalized distance

$$\delta_{ij} = \frac{1}{D_z} \sum_{d=1}^{D_z} \frac{(\mu_{i,d} - \mu_{j,d})^2}{v_{i,d} + v_{j,d} + \varepsilon}. \tag{4}$$

Define compatibility and the different-original positive weight as

$$C_{ij} = \frac{1}{1 + \delta_{ij}/\gamma}, \qquad w^+_{ij} = 1 - \lambda(e)(1 - C_{ij}), \quad j \in \mathcal{P}_i \text{ with } o_j \neq o_i. \tag{5}$$

Here $\gamma > 0$ sets the compatibility scale and $\lambda(e)$ is the positive curriculum strength. High compatibility gives a weight near one; low compatibility reduces the influence of that positive, especially when $\lambda(e)$ is large. For two views of the same original video, $w^+_{ij} = 1$.

3 Role of the negative weights

The negative weights α_{ik} appear in the denominator Z_i, rather than defining a second loss. A larger α_{ik} makes negative k compete more strongly against the positive samples. Define

$$p_{ik}^{-} = \frac{\exp(-\delta_{ik}/\tau_{\text{neg}})}{\sum_{j\in\mathcal{N}_i}\exp(-\delta_{ij}/\tau_{\text{neg}})}, \qquad k \in \mathcal{N}_i, \tag{6}$$

$$\alpha_{ik} = |\mathcal{N}_i| p_{ik}^{-}, \qquad \sum_{k\in\mathcal{N}_i}\alpha_{ik} = |\mathcal{N}_i|. \tag{7}$$

This preserves the total negative weight per anchor. A high negative temperature τ_{neg} distributes weights broadly; a lower value concentrates weight on negatives with smaller uncertainty-normalized distances. The negative selection temperature τ_{neg} is distinct from the semantic contrastive temperature τ.

4 Why there is only one contrastive objective

The positive mechanism modifies the coefficients used to average positive log-probabilities in ℓ_i. The negative mechanism modifies the partition function Z_i within those same probabilities. Both therefore act on one supervised contrastive objective; there is no separate positive-contrastive loss and negative-contrastive loss. The *full training objective*, however, also includes a distinct uncertainty-training term.

5 Why gradients are stopped through pair weights

Although δ_{ij} depends on both $\boldsymbol{\mu}$ and $\boldsymbol{v}$, the implementation forms pair weights from

$$\delta_{ij}^{\text{weight}} = \text{SG}(\delta_{ij}), \tag{8}$$

where SG denotes stop-gradient. Consequently, contrastive optimization cannot directly manipulate the variance estimator through the weighting rules. This blocks a shortcut in which the network increases or decreases variance merely to change which pairs are emphasized. Contrastive gradients still update semantic means through the cosine-similarity logits. The variance branch consequently needs its own learning signal.

6 Temporal uncertainty objective: definitions

At temporal position t and latent dimension d, the Gaussian head predicts a mean $\mu_{i,t,d}$ and a positive variance $v_{i,t,d}$. The video-level mean $\mu_{i,d}$ is the average over valid temporal positions. Define the residual relative to a detached center:

$$r_{i,t,d} = \mu_{i,t,d} - \text{SG}(\mu_{i,d}). \tag{9}$$

The stated heteroscedastic objective is

$$\boxed{L_{\text{unc}} = \frac{1}{BTD_z}\sum_{i=1}^{B}\sum_{t=1}^{T}\sum_{d=1}^{D_z}\left[\frac{r_{i,t,d}^2}{v_{i,t,d}} + \log v_{i,t,d}\right].} \tag{10}$$

The squared-residual-over-variance term penalizes predicting too little variance for large temporal deviations; the logarithmic term penalizes unnecessarily large variance. This is a Gaussian heteroscedastic negative-log-likelihood-like objective, omitting the conventional factor $1/2$ and additive constant.

7 Derivation of the variance optimum

For a fixed residual r and $v > 0$, consider

$$f(v) = \frac{r^2}{v} + \log v. \tag{11}$$

Differentiating yields

$$\frac{\mathrm{d}f}{\mathrm{d}v} = -\frac{r^2}{v^2} + \frac{1}{v} = \frac{v - r^2}{v^2}. \tag{12}$$

For $r \neq 0$, the stationary point is $v = r^2$, which minimizes the objective. The variance head is thus encouraged to match the squared magnitude of temporal residuals. If $r = 0$, the unconstrained objective favors $v \to 0$; the implementation imposes a positive variance floor. This does not by itself guarantee calibration to true predictive uncertainty.

8 Critical distinction: detaching the residual

Detaching only the center in

$$r_{i,t,d} = \mu_{i,t,d} - \mathrm{SG}(\mu_{i,d}) \tag{13}$$

does not prevent L_{unc} from updating $\mu_{i,t,d}$. In the supplied Python implementation, the residual is also detached before evaluating the uncertainty objective:

$$\widetilde{r}_{i,t,d} = \mathrm{SG}\big(\mu_{i,t,d} - \mathrm{SG}(\mu_{i,d})\big). \tag{14}$$

The implemented objective therefore uses $\widetilde{r}^2_{i,t,d}/v_{i,t,d}$, so this loss primarily trains the variance head without directly forcing temporal semantic means toward their center. Removing the outer stop-gradient would define a different optimization design.

9 Handling padded temporal positions

The unmasked expression assumes every one of the BT temporal positions is valid. In the implementation, $m_{it} \in \{0, 1\}$ indicates validity. The masked form is

$$\boxed{L_{\mathrm{unc,masked}} = \frac{1}{D_z \sum_{i,t} m_{it}} \sum_{i,t,d} m_{it} \left[\frac{\widetilde{r}^2_{i,t,d}}{v_{i,t,d}} + \log v_{i,t,d} \right].} \tag{15}$$

The video mean used in the residual also averages only valid positions. The code clamps the normalization denominator to avoid division by zero; training data should still contain valid positions.

10 Combined objective and gradient pathways

The final objective is

$$\boxed{L_{\mathrm{total}} = L_{\mathrm{contrast}} + \beta_{\mathrm{unc}} L_{\mathrm{unc}}.} \tag{16}$$

The contrastive component updates semantic representations through cosine-similarity logits while treating uncertainty-based pair weights as fixed statistics during each gradient step. The uncertainty component updates the predicted temporal variances using detached temporal residuals. The parameter β_{unc} controls its relative contribution; the supplied implementation uses $\beta_{\mathrm{unc}} = 0.01$ by default.

Video-level variance contains both the average predicted temporal variance and temporal variability of means. This combined variance determines pairwise uncertainty-normalized distances, but its path into contrastive weights is detached. The explicit learning signal to the variance head in this design is therefore the uncertainty objective.

11 Worked example

Suppose an anchor has two positives with weights $w_1^+ = 1.0$ and $w_2^+ = 0.3$, and log-probabilities -0.4 and -1.2. Then

$$\ell_i = -\frac{1.0(-0.4) + 0.3(-1.2)}{1.0 + 0.3} = \frac{0.76}{1.3} \approx 0.585. \tag{17}$$

Without positive weighting, the average loss would be 0.8. The lower-weight positive therefore has less influence. Independently, increasing the weight of a hard negative enlarges Z_i, reduces the positive log-probabilities, and increases the contrastive penalty. These are two adjustments to the same ℓ_i.

12 What the objective establishes and what requires experiments

The equations specify a coherent way to combine semantic alignment, uncertainty-aware pair weighting, and a separate variance-learning signal. They do not, on their own, establish that the variance is calibrated, that selected negatives are semantically hard, or that downstream representations improve. These claims require empirical comparisons against unweighted supervised contrastive learning, positive-only weighting, negative-only weighting, and ablations with and without the uncertainty loss and gradient stopping.

Implementation alignment. The explanation follows the supplied `TemporalGaussianHead`, `build_positive_weights`, `build_negative_weights`, `temporal_uncertainty_loss`, and `StableUncertaint` components.